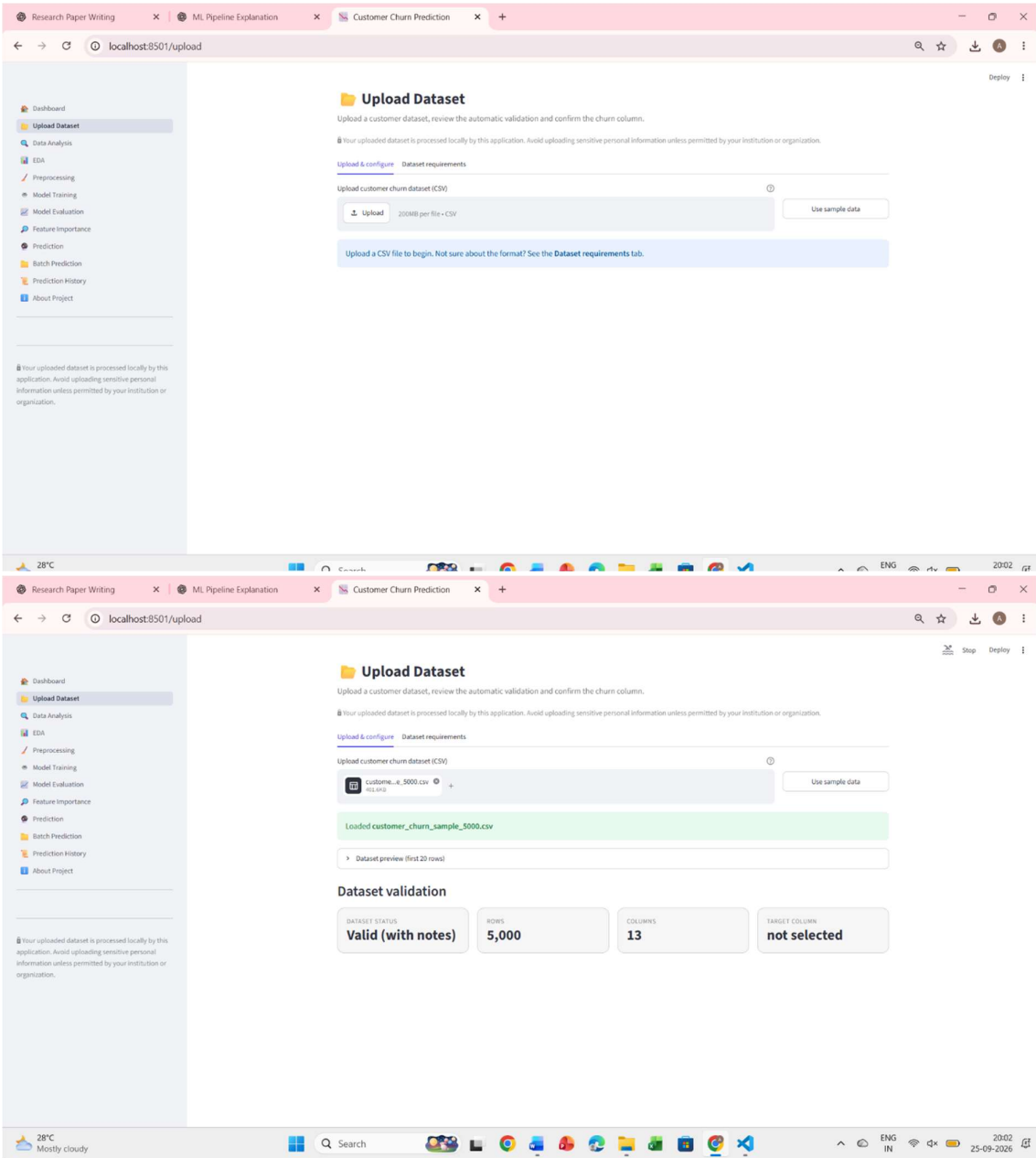
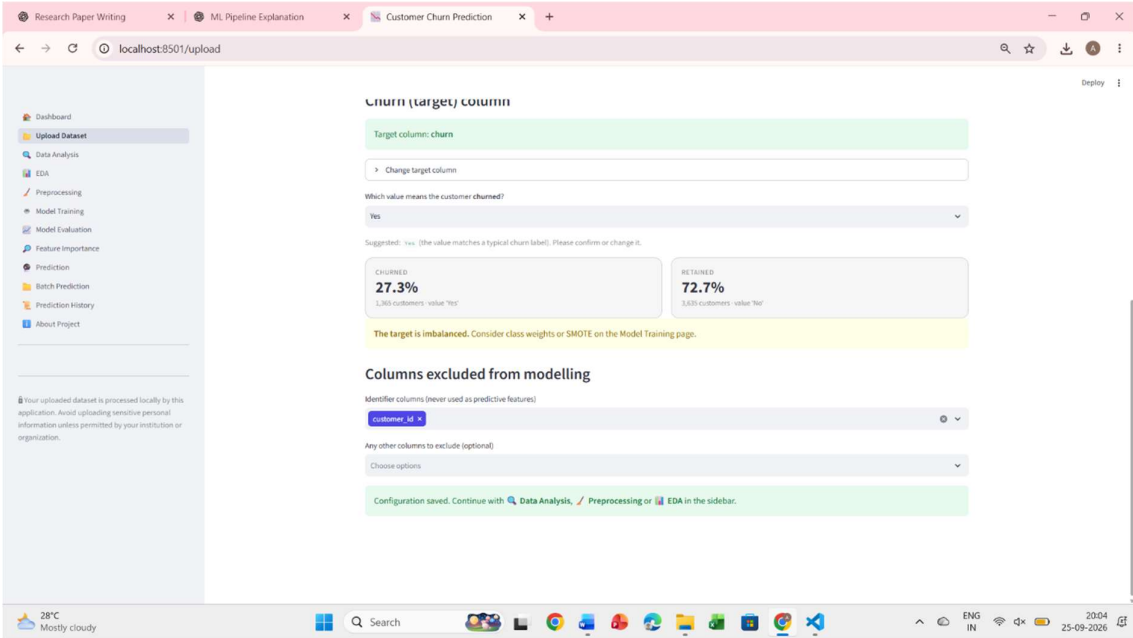
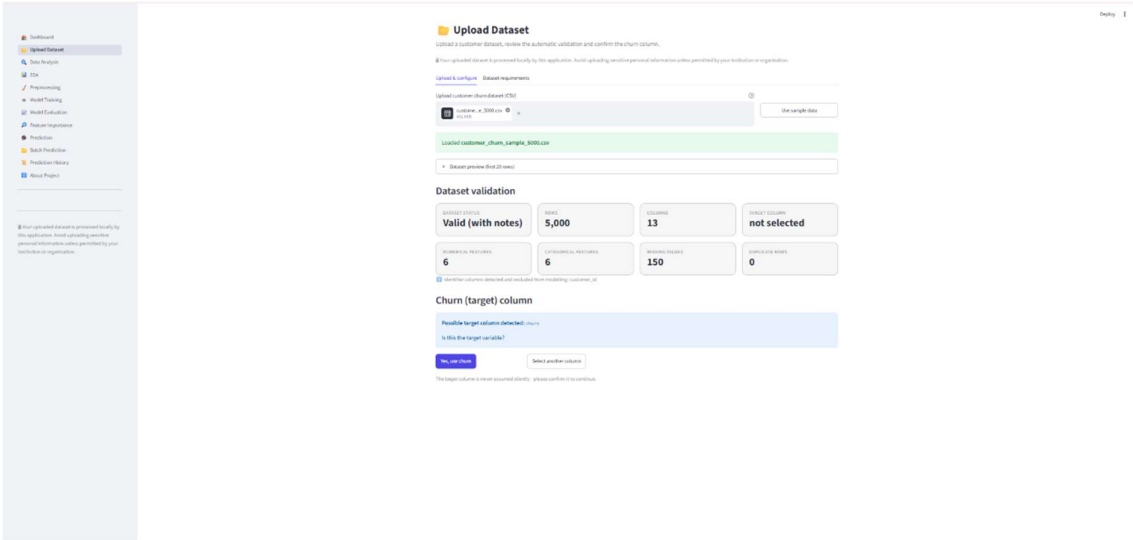






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About Project

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Data Analysis

Automatic analysis of the uploaded dataset's structure and quality.

DATASET STATUS

Valid (with notes)

ROWS

5,000

COLUMNS

13

TARGET COLUMN

churn
churn = "Yes"

NUMERICAL FEATURES

6

CATEGORICAL FEATURES

5

MISSING VALUES

150
in 3 columns

DUPLICATE ROWS

0
0 ignoring ID columns

Column profile

Missing values

Target distribution

Quality notes

Download

column	raw dtype	inferred role	missing	missing %	unique	example
customer_id	str	identifier	0	0	5000	C00001
age	int64	numeric	0	0	57	23
gender	str	categorical	50	1	2	Male
region	str	categorical	50	1	5	South
tenure	int64	numeric	0	0	72	6
monthly_charges	float64	numeric	0	0	4042	187.88
total_charges	float64	numeric	0	0	4979	594.95
contract	str	categorical	0	0	3	Month-to-month
internet_service	str	categorical	0	0	3	Fiber optic
support_tickets	int64	numeric	0	0	19	5

Column profile (CSV)

28°C
Mostly cloudy

Search

ENG IN

20:04
25-09-2026

localhost:8501/eda

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TOTAL CUSTOMERS

5,000

CHURN RATE

27.3%

TOTAL FEATURES

10

NUMERICAL / CATEGORICAL

5 / 5

MISSING VALUES

150
in the cleaned data

DUPLICATE RECORDS

0
after cleaning

CHURNED CUSTOMERS

1,365

RETAINED CUSTOMERS

3,635

Churn distribution

Numerical

Categorical

Correlation

Key visualizations

Feature explorer

Churn count

Churn percentage

Churned

Retained

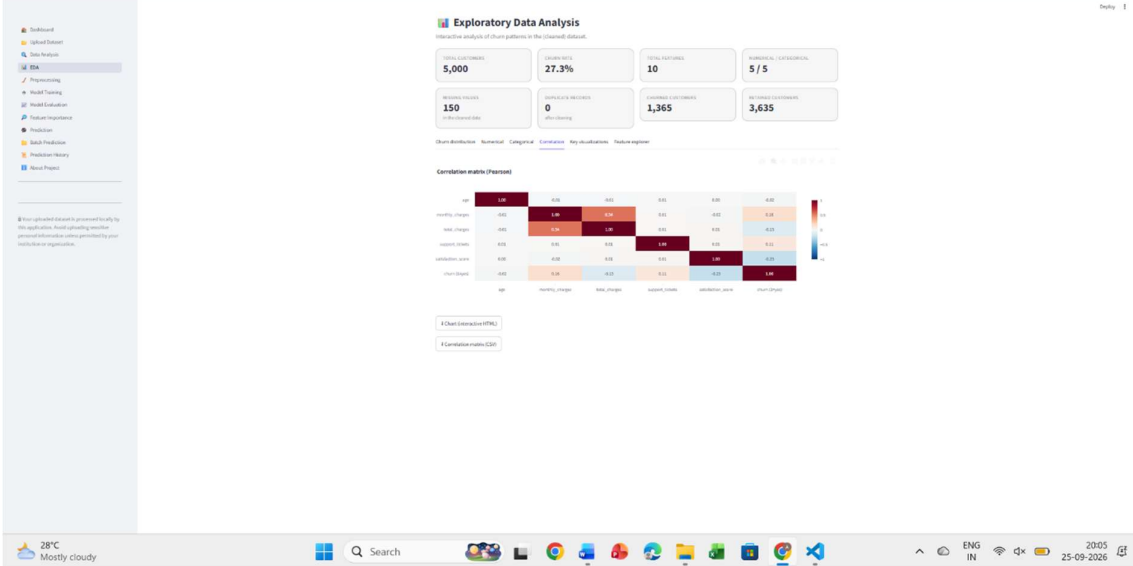
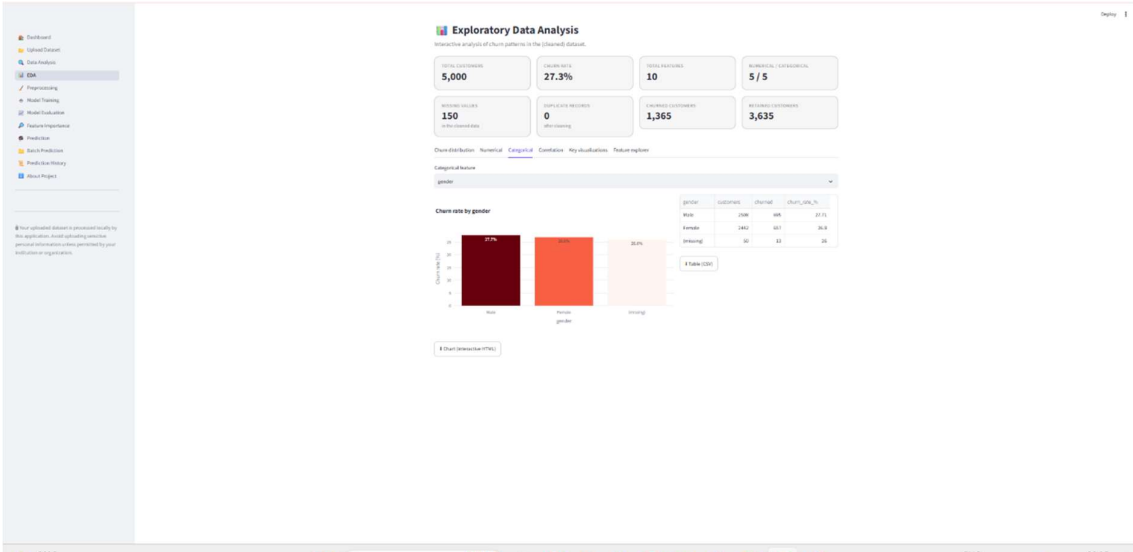
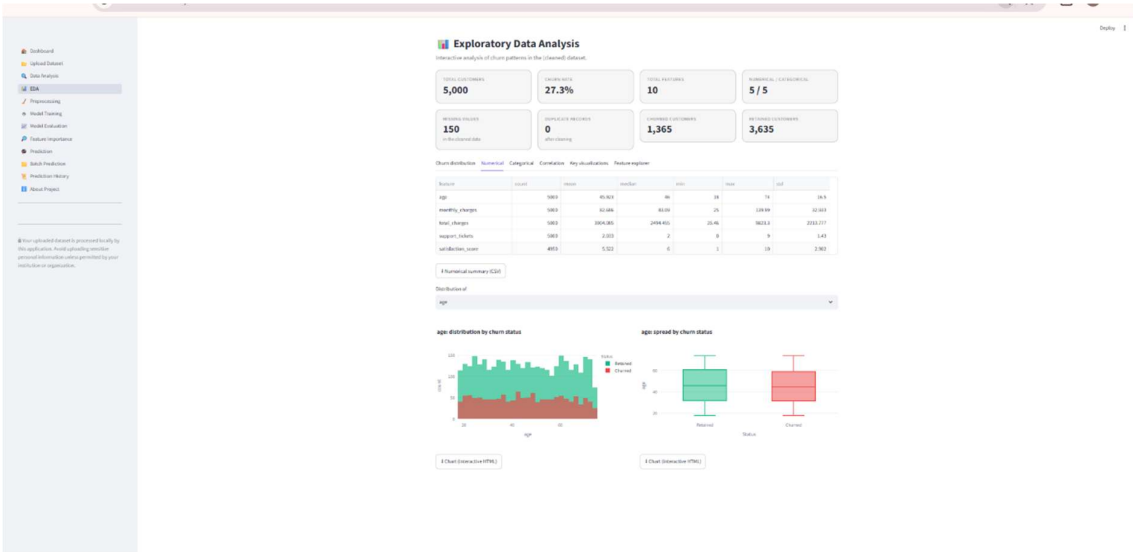
Churned 27.3%

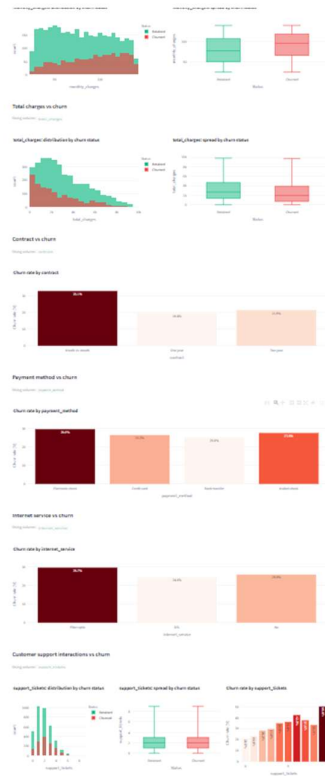
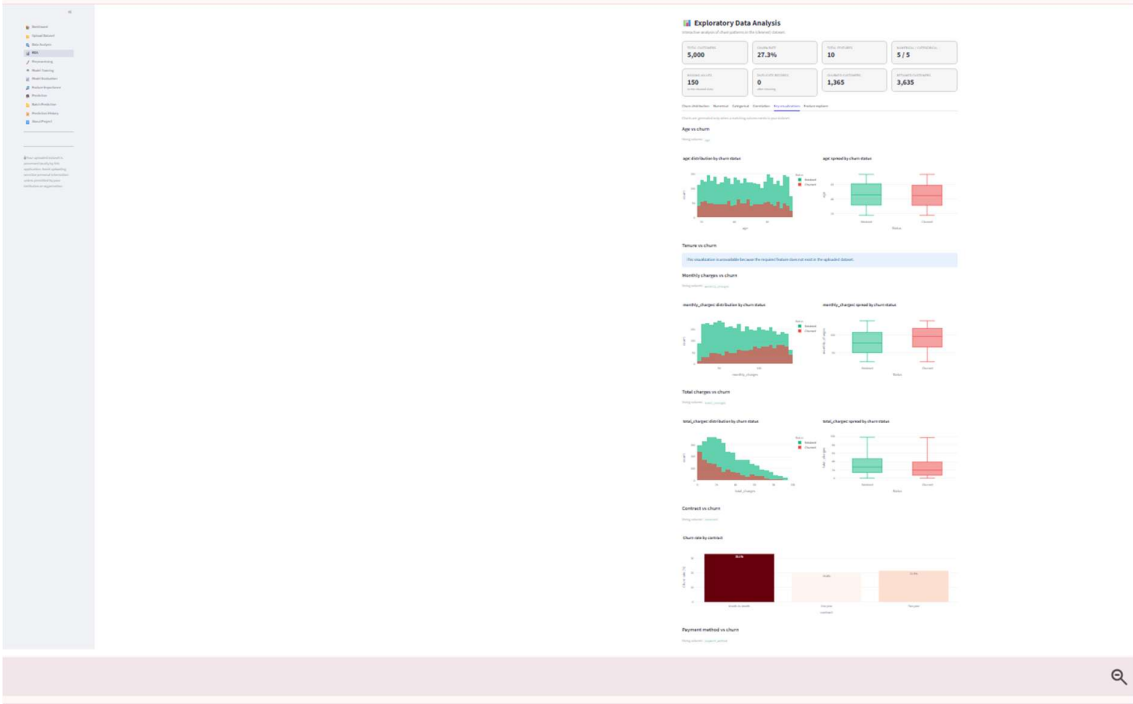
Retained 72.7%

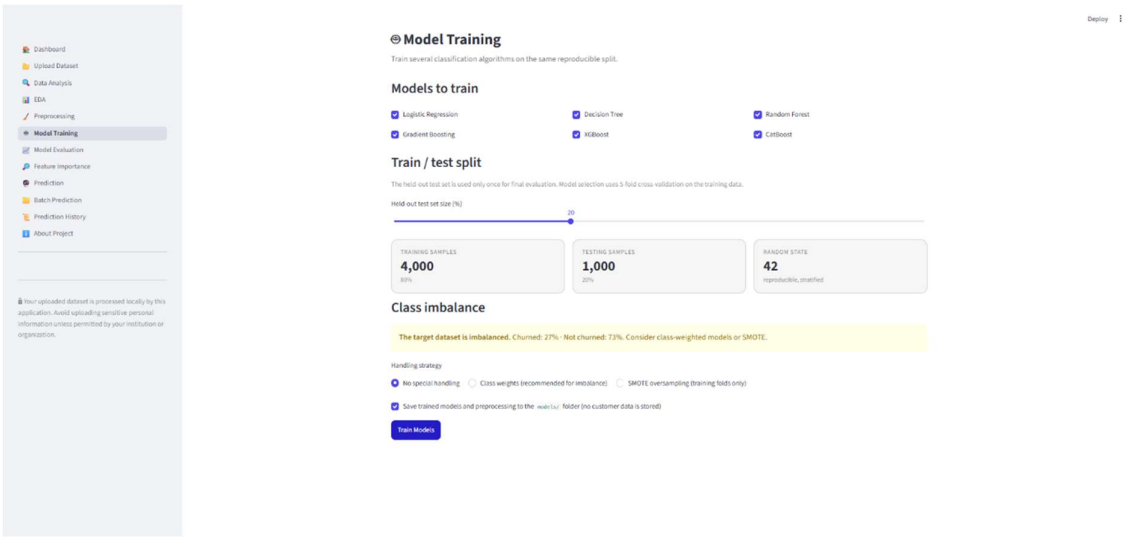
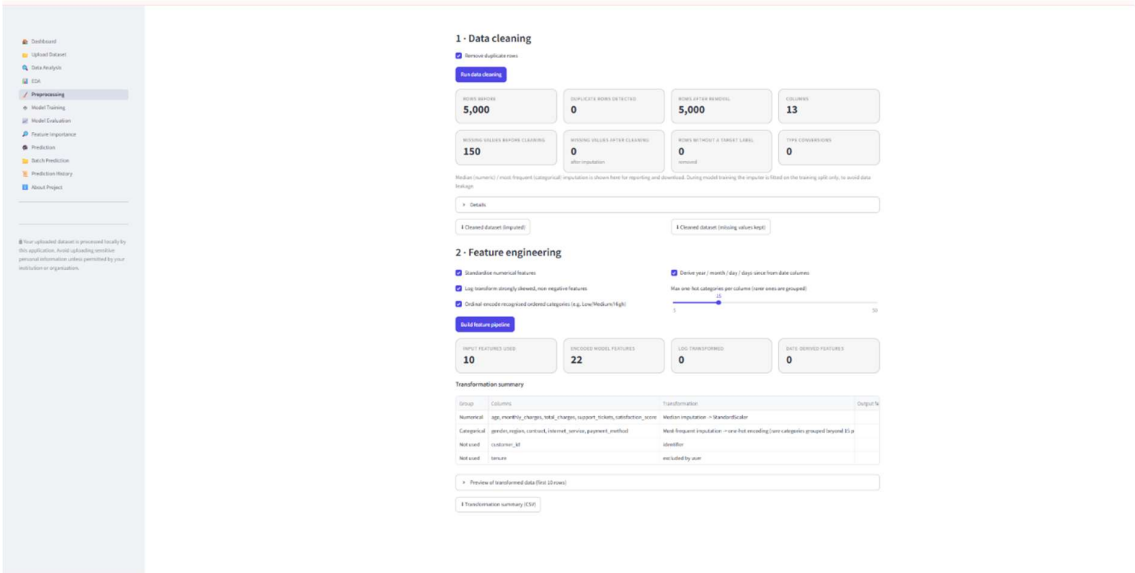
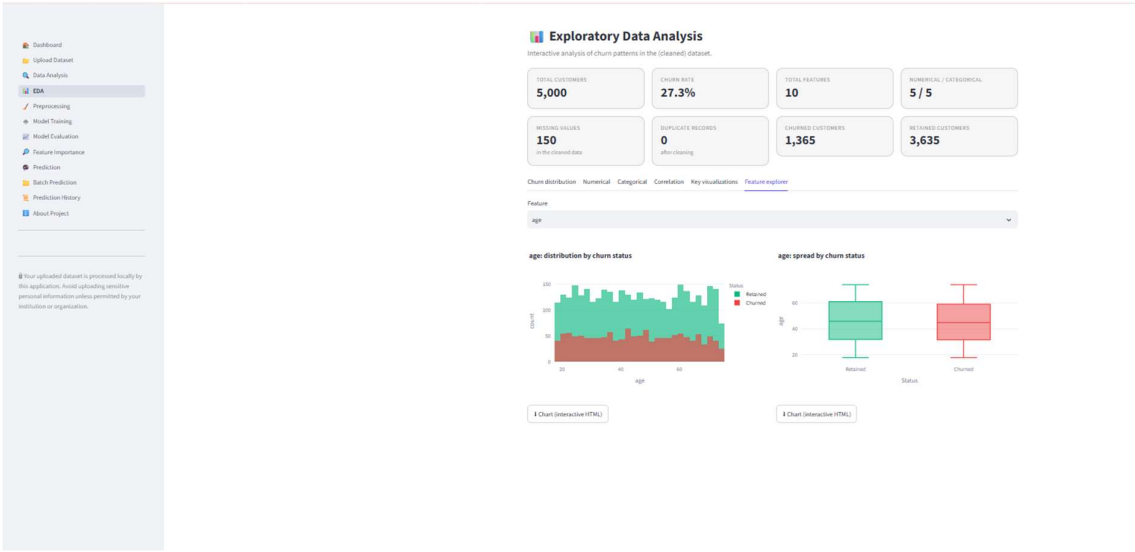
Chart (Interactive HTML)

Chart (Interactive HTML)

Churned: 27.3% - Retained: 72.7% (calculated from the data)







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4,000
80%

1,000
20%

42
reproducibility, stratified

Class imbalance

The target dataset is imbalanced. Churned: 27%. Not churned: 73%. Consider class-weighted models or SMOTE.

Handling strategy

☐ No special handling ☒ Class weights (recommended for imbalance) ☐ SMOTE oversampling (training folds only)

☒ Save trained models and preprocessing to the `workspace` folder (no customer data is stored)

[Train Models](#)

Trained 6 model(s) on 4,000 samples, evaluated once on 1,000 held-out samples. Prediction model selected by 5-fold CV ROC-AUC: **Logistic Regression**.

Training summary

Model	Status	Train time (s)	Details
Logistic Regression	trained		0.53
Decision Tree	trained		0.65
Random Forest	trained		6.16
Gradient Boosting	trained		9.06
XGBoost	trained		7.41
CatBoost	trained		11.21

[Go to Model Evaluation to compare the models.](#)

Saved to `workspace/preprocessor.pkl`, `feature_schema.json`, `one_hot.pkl` per model.

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Model Evaluation

Compare models on the held-out test set. All numbers are computed from your data.

Model comparison Per model evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	CV ROC-AUC	CV ROC-AUC Std	Train-time (s)
Logistic Regression	0.696	0.4511	0.63	0.5325	0.7635	0.5447	0.7706	0.0163	0.53
Decision Tree	0.672	0.433	0.663	0.5239	0.733	0.4766	0.7079	0.0105	0.65
Random Forest	0.744	0.5221	0.3297	0.4128	0.7278	0.525	0.7451	0.0244	6.16
Gradient Boosting	0.714	0.482	0.6374	0.5489	0.7625	0.5046	0.7543	0.0195	9.06
XGBoost	0.713	0.477	0.6081	0.5345	0.7443	0.541	0.7447	0.0207	7.41
CatBoost	0.705	0.4676	0.5824	0.5188	0.7484	0.5417	0.7505	0.018	11.21

Model comparison (CSV)

SELECTION METHOD

5-fold CV
training data only

CV-SELECTED MODEL

Logistic Regression
ROC-AUC: 0.7706

The model shown as selected was chosen using 5-fold cross-validation on the training partition. The held-out test set is reserved for final, unbiased performance reporting.

Test set metric to compare models

ROC-AUC

Highest ROC-AUC on this held-out test set: **Gradient Boosting (0.7635)**. This is descriptive only and is not used for automatic model selection.

ROC-AUC by model

All metrics

Dashboard

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Test set metric to compare models

F1

Highest F1 on this held-out test set: **Gradient Boosting (0.5489)**. This is descriptive only and is not used for automatic model selection.

F1 by model

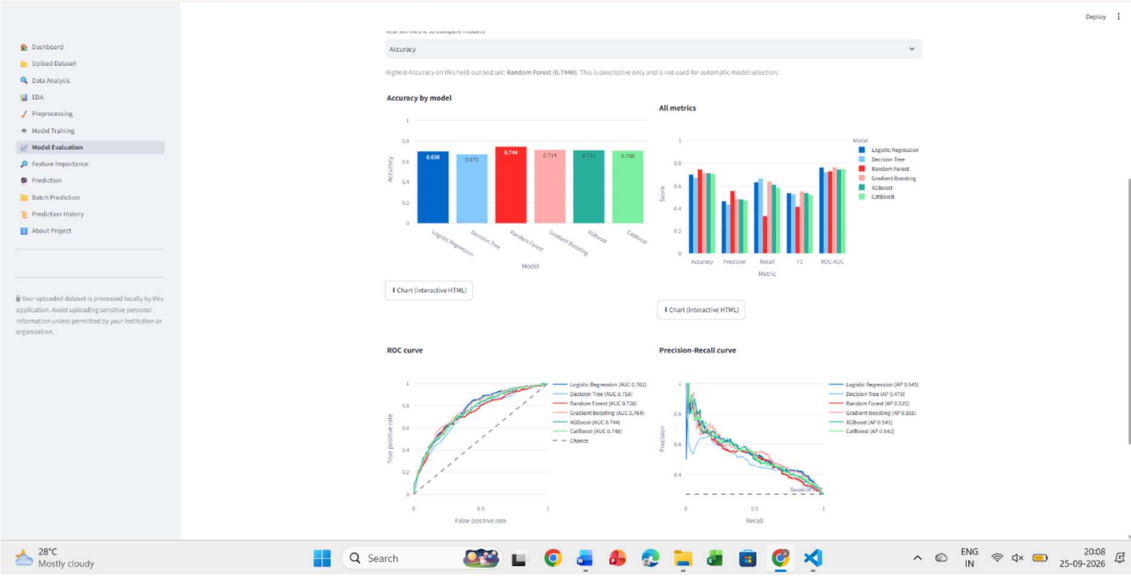
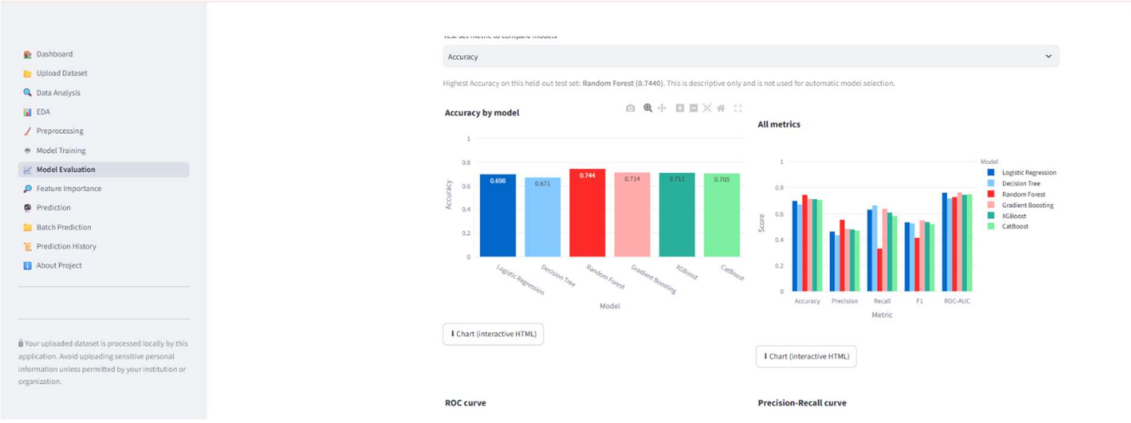
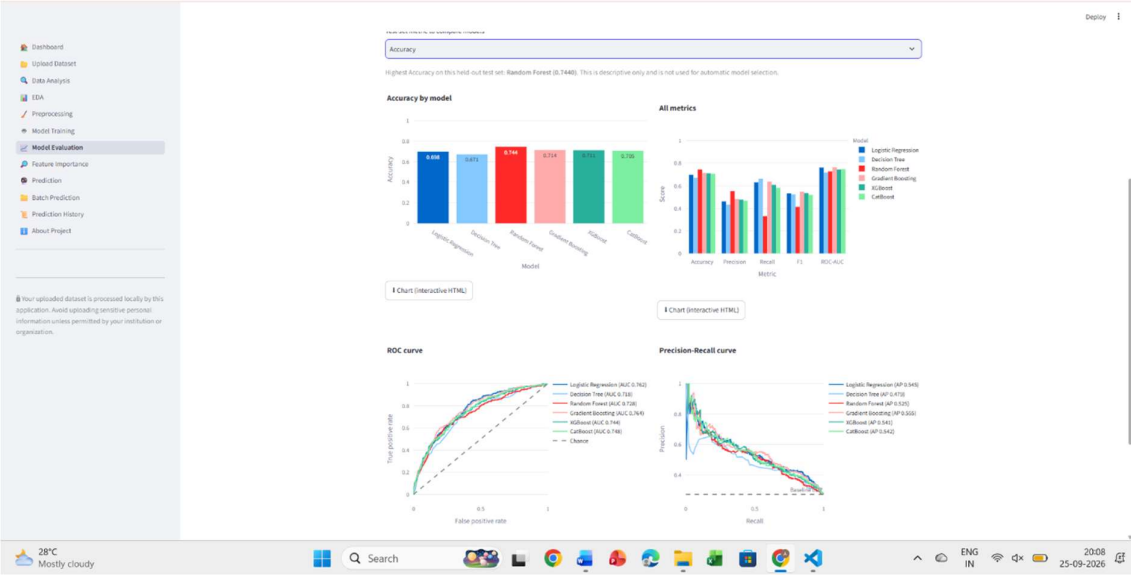
All metrics

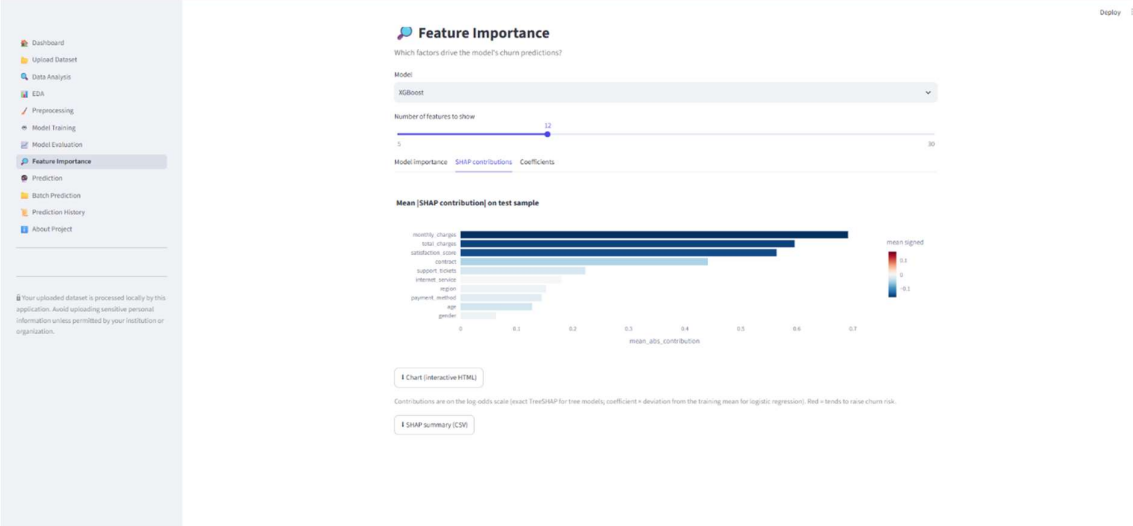
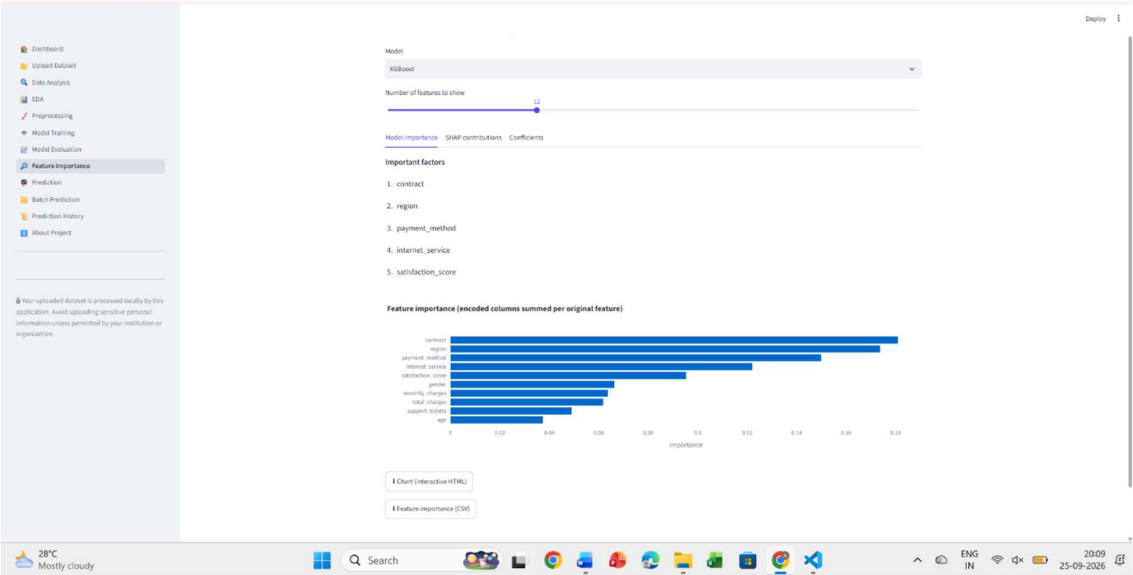
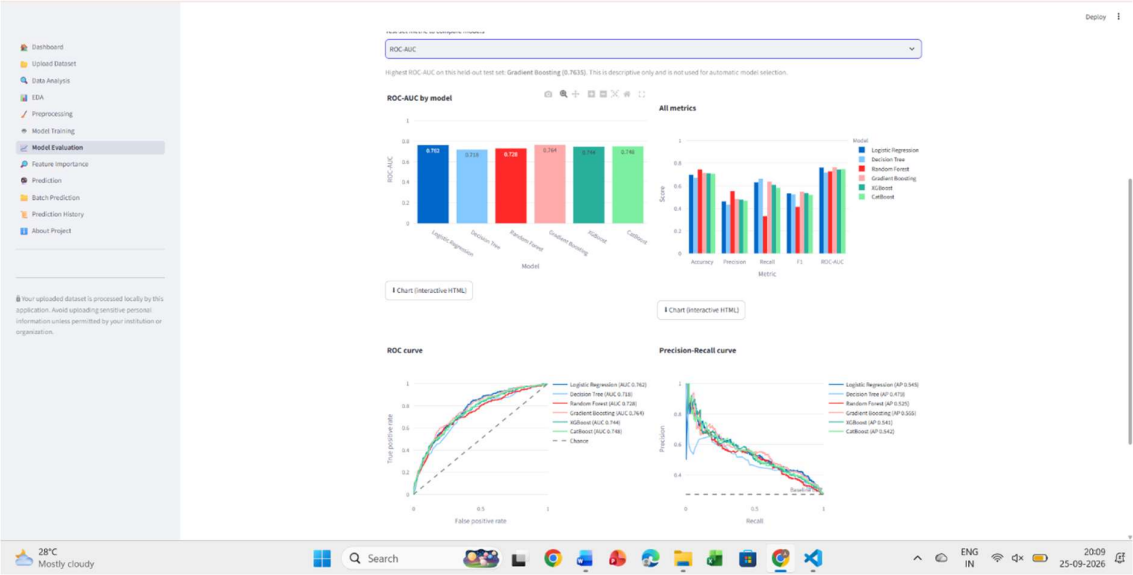
Chart (interactive HTML)

Chart (interactive HTML)

ROC curve

Precision-Recall curve





Dashboard

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About Project

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Customer Prediction

Enter one customer's details to estimate their churn risk.

Use a previously saved model instead

Only load model files you created yourself (pickle files are pickles).

Saved model

decision_tree.joblib

Load saved model

Switch back to this session's model

Model: Decision Tree (saved model). Churn = 'Yes'

Risk and decision thresholds

Low = Medium at0.30Medium = High at0.60High = Very High at0.80Churn decision threshold0.50

Low = 30% ≤ Medium = 60% ≤ High = 80% ≤ Very High

Customer details

age46genderMaleregionNorth

monthly_charges83.17total_charges2470.24contractMonth to month

internet_serviceFiber opticupport_tickets2satisfaction_score6

payment_methodElectronic check

Predict Churn

28°C

Mostly cloudy

Search

ENG IN

20:10

25-09-2026

Dashboard

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About Project

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prediction_method

Electronic check

Predict Churn

Prediction Result

Customer is predicted as:

NOT LIKELY TO CHURN

Churn probability: 38.80% · Non-churn probability: 61.20%

Risk level: NEEDS ATTENTION

The probability is a model estimate based on patterns in the uploaded data, not a guarantee of what this customer will do.

38.8%

Prediction Explanation

The model's prediction was influenced by:

- monthly_charges = 83.17 lowers churn risk
- satisfaction_score = 6 lowers churn risk
- contract = Month-to-month raises churn risk
- total_charges = 2470.24 lowers churn risk
- support_tickets = 2 raises churn risk
- age = 46 lowers churn risk

Feature contributions for this customer (log odds)

contract

support_tickets

internet_service

region

monthly_charges

total_charges

payment_method

gender

contribution

offset

Month-to-month

Needs Attention

Contributions are computed from the trained model (SDRP) using positive values push the prediction toward churn.

Saved to the session's Prediction History.

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About Project

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Batch Prediction

Upload a CSV of customers (without the churn column) and score them all.

Your uploaded dataset is processed locally by this application. Avoid uploading sensitive personal information unless permitted by your institution or organization.

Model: Decision Tree (saved model). Required columns: age , gender , region , monthly_charges , total_charges , contract , internet_service , support_tickets , satisfaction_score , payment_method

Risk and decision thresholds

Low = Medium at0.30Medium = High at0.60High = Very High at0.80Churn decision threshold0.50

Low = 30% ≤ Medium = 60% ≤ High = 80% ≤ Very High

Upload customer prediction file (CSV)

cleaned_dataset.csv · 817.2 KB

Use sample file

Run Batch Prediction

Scored 5,000 customers · 2,148 predicted to churn (43.0%).

customer_id	prediction	churn_probability	risk_level
C00001	Yes	0.8099	Very High
C00002	No	0.3923	Medium
C00003	No	0.4509	Medium
C00004	No	0.2405	Low
C00005	No	0.0499	Low
C00006	Yes	0.7105	High
C00007	No	0.3836	Medium

28°C

Mostly cloudy

Search

ENG IN

20:11

25-09-2026

Dashboard

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About the Project

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

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Abstract & objectives Methodology Technology Limitations

Dataset Upload

Data Validation

Data Cleaning

EDA

Feature Engineering

Preprocessing

Train/Test Split

Classification Models

Model Evaluation

Model Comparison

Churn Prediction

Visualization & Report

Every model shares one scikit-learn preprocessing pipeline that is fitted on the training split only (no leakage), stored with the model together with the feature schema, and reused unchanged at prediction time.

Deploy

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About the Project

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

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Abstract & objectives Methodology Technology Limitations

- Python, Streamlit (UI), Plotly (interactive charts)
- pandas / NumPy / scikit-learn (pipelines, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting)
- XGBoost, CatBoost (boosted trees), imbalanced-learn (SMOTE), SHAP (tree explanations), joblib (persistence)

Deploy

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About the Project

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

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Abstract & objectives Methodology Technology Limitations

- Predictions are statistical estimates from the uploaded data, not guarantees; quality depends on data quality and size.
- Model selection uses 5-fold cross-validation on the training partition; the held-out test set is reserved for final reporting.
- Explanations show association within the model, not causation.
- SMOTE interpolates over encoded features, so synthetic one-hot values are fractional; class weights are the safer default.
- The target must be binary; multi-class targets are rejected with guidance.
- Data lives in the browser session only; saved model files (`models/`) contain no customer rows but should be treated as pickles - load only your own.

Deploy